RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Thursday, 17 September 2026, 10:16 AM
State	Finished
Completed on	Sunday, 27 September 2026, 10:05 AM
Time taken	9 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs to run a distance to burn out his calories.

If he has eaten i burgers with c calories each, then he has to run at least $3^i * c$ kilometers to burn out the calories. For example, if he ate 3

burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are $(3^0 * 1) + (3^1 * 3) + (3^2 * 2) = 1 + 9 + 18 = 28$.

But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine the minimum distance he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy approach to solve the problem.

Input Format

First Line contains the number of burgers

Second line contains calories of each burger which is n space-separate integers

Output Format

Print: Minimum number of kilometers needed to run to burn out the calories

Sample Input

```
3
5 10 7
```

Sample Output

```
76
```

For example:

Test	Input	Result
Test Case 1	3 1 3 2	18

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include<math.h>

void merge(int arr[], int low, int mid, int high) {
    int temp[1000];
    int i = low, j = mid + 1, k = 0;

    while (i <= mid && j <= high) {
        if (arr[i] >= arr[j])
            temp[k++] = arr[i++];
        else
            temp[k++] = arr[j++];
    }

    while (i <= mid)
        temp[k++] = arr[i++];

    while (j <= high)
        temp[k++] = arr[j++];

    for (i = low, k = 0; i <= high; i++, k++)
        arr[i] = temp[k];
}

void mergeSort(int arr[], int low, int high) {
    if (low < high) {
        int mid = (low + high) / 2;

        mergeSort(arr, low, mid);
        mergeSort(arr, mid + 1, high);

        merge(arr, low, mid, high);
    }
}

int main() {
    int n;
    scanf("%d", &n);
```

```
39
40      int arr[n];
41
42      for (int i = 0; i < n; i++)
43          scanf("%d", &arr[i]);
44
45      mergeSort(arr, 0, n - 1);
46
47      long long sum = 0;
48
49
50      for (int i = 0; i < n; i++) {
51          sum += arr[i] * pow(n,i);
52
```

	Test	Input	Expected	Got	
✔	Test Case 1	3 1 3 2	18	18	✔
✔	Test Case 2	4 7 4 9 6	389	389	✔
✔	Test Case 3	3 5 10 7	76	76	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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